

# Operation Manual – Metal Proximity Counting Device

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## 1. Overview

The Metal Proximity Counting Device is an AC-powered, Wi-Fi-enabled system designed to count metallic objects passing near its proximity sensor. The device supports MQTT communication for data transmission to a server or cloud application.

It features two modes of operation:

**Counting Mode** – normal operation for metal detection and counting.

**Setup Mode** – configuration of Wi-Fi and MQTT parameters through an onboard captive portal.

## 2. Specifications

| <u>Parameter</u> | <u>Description</u>                 |
|------------------|------------------------------------|
| Power Supply     | AC 230 V                           |
| Sensor Type      | Metal proximity sensor             |
| Communication    | Wi-Fi (2.4 GHz)                    |
| Protocol         | MQTT                               |
| User Interface   | One setup button, one reset button |
| Indicators       | Bi-color LED (Green/Red)           |
| Enclosure        | Industrial-grade ABS               |



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## 3. Functional Description

### 3.1 Operating Modes

The device operates in two distinct modes:

#### A. Counting Mode (Normal Operation)

Default mode on power-up.

The device continuously monitors the proximity sensor.

Each detection event increments the internal count value by one.

The count value is transmitted to the configured MQTT broker once it reaches the threshold count set by the user.

A reset button can reset the count to zero at any time.

#### B. Setup Mode (Configuration Mode)

Accessed by long pressing the setup key (for approximately 6 seconds).

The device enters Access Point (AP) mode and creates a Wi-Fi captive portal for configuration.

Through the captive portal, the user can configure:

Wi-Fi SSID and password

MQTT broker address

MQTT topic name

Threshold count (number of detections before sending MQTT message)

Device ID (for identification on MQTT network)

After configuration is saved, the device reboots and returns to Counting Mode.

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## 4. Indicator LED Status

| <u>LED Color / Behavior</u> | <u>Meaning</u>                                |
|-----------------------------|-----------------------------------------------|
| <b>Green</b> (steady)       | Normal operation; Wi-Fi connected             |
| <b>Red</b> (steady)         | Wi-Fi not connected                           |
| <b>Red</b> (blinking)       | Setup Mode active (AP captive portal running) |

## 5. Buttons and Their Functions

| <u>Button</u> | <u>Action</u>                  | <u>Function</u>                                    |
|---------------|--------------------------------|----------------------------------------------------|
| Setup Button  | Short press                    | No action in counting mode                         |
|               | Long press ( $\geq 6$ seconds) | Enter Setup Mode (AP mode with captive portal)     |
| Reset Button  | Short press                    | reset the count value to 0 (in counting mode only) |

## 6. Configuration via Captive Portal

Long press the multifunction button until the LED begins blinking red.

On a Wi-Fi-enabled device (phone or PC), connect to the hotspot named:

**METAL\_CNT\_IoT\_AP**                      password: 12345678

Once connected, a configuration page will open automatically. If not, navigate manually to <http://192.168.4.1> in a web browser.

Fill in the following fields:

Wi-Fi SSID

Wi-Fi Password

MQTT Broker Address

MQTT Topic

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Count to send (Threshold Count)

Device ID

Click Submit.

The device will restart, connect to the configured Wi-Fi network, and enter Counting Mode.

## Enter Metal Count Information

Scan for Wi-Fi

Select a network...

▼

Submit

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## 7. MQTT Communication

The device publishes messages to the specified MQTT topic when the count reaches the configured threshold.

Example message format:

```
{  
  "device_id": "DEV001",  
  "count": 50,  
  "timestamp": "2025-11-11T14:32:00Z"  
}
```



## 8. Power and Installation

Ensure power source is AC 230 V  $\pm 10\%$ .

Connect the proximity sensor and ensure correct alignment with the target object.

Mount the device securely in a dry, vibration-free environment.

Ensure Wi-Fi signal strength is adequate for stable operation.

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## 9. Troubleshooting

| <u>Issue</u>                  | <u>Possible Cause</u>                         | <u>Solution</u>                         |
|-------------------------------|-----------------------------------------------|-----------------------------------------|
| LED steady red                | Wi-Fi not configured or disconnected          | Enter Setup Mode and reconfigure Wi-Fi  |
| LED blinking red continuously | Device stuck in Setup Mode                    | Save configuration and reboot           |
| No MQTT messages              | Incorrect MQTT settings or broker unreachable | Recheck MQTT broker address and topic   |
| Count not incrementing        | Sensor not detecting                          | Check sensor alignment and metal target |

## 10. Safety Precautions

Only qualified personnel should handle AC connections.

Disconnect power before servicing.

Do not expose to water or conductive dust.

Ensure correct grounding of the device.

## 11. Technical Support

For technical assistance, contact your system integrator or email: [r\\_d\\_lab@siliconharvest.net](mailto:r_d_lab@siliconharvest.net)